

BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY

Department of Computer Science and Engineering

L-2/T-2 CSE 219: Signals and Linear Systems

Time: 25 minutes

Marks: 20

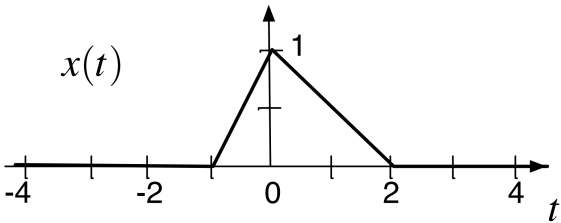
Student Name: _____

Student No: _____

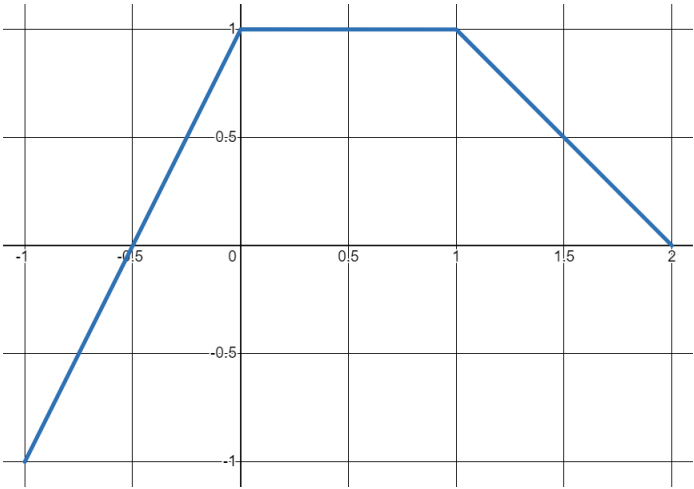
1. The graph of a signal $x(t)$ is given below. Please draw the graphs of the two given signals. (2+2=4 Marks)

1. $x(-2t + 1)$

2. $x(-\frac{t}{2} + 2)$



2. The graph of a function $y(t)$ is given below, where x-axis of the graph denotes time t and y-axis denotes $y(t)$. Please determine the even-odd decomposition of this function and draw the graphs of the both functions. You are free to assume any reasonable intervals within this piecewise function. (8 Marks)



3. Suppose $x[n] = (3 + 4j)e^{2nj}$ is a complex exponential discrete-time signal. Please determine the sinusoidal signals constituting the real and the imaginary parts of $x[n]$. Kindly show whether these two signals are periodic or not. You have to show the intermediate steps and justify your answer. You need not draw any graphs. (6 Marks)

4. The output $y(t)$ of a system for input $x(t)$ is given below. Please determine whether this system is a linear and time-invariant (LTI) system. Kindly show the intermediate steps. (2 Marks)

• $y(t) = \sin(t) + x(t)$